



Procedural generation of lens soiling data via physics-based simulation

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Abstract

Detecting soiling on optical camera lenses using deep learning-based algorithms requires substantial datasets, which are challenging to acquire in large volumes at a low cost. Existing data generation techniques have limitations in fidelity, physical interpretability, and artistic input. We present a novel physics-based method for procedural generation of lens soiling data that seamlessly integrates with deep learning training pipelines. Our approach simulates the imaging process of lens soiling through physical optics and dissecting it into discrete stages. By harnessing fundamental image processing functions from OpenCV, we synthesize a variety of effects, including mud stains, water stains, water droplet refraction, and lens flare. This methodology ensures the effectiveness, interpretability, and high annotation accuracy of the generated data. The algorithms utilized are grounded in fundamental image processing functions, making them suitable for integration into standard deep learning training pipelines. Experiments demonstrate the efficacy of our generated data in the task of image classification on the real-world public dataset. In the experiments of soiling semantic segmentation, we also achieve improved results by adjusting the categories. Thereby, we validate its practical utility and potential for enhancing the performance of lens soiling detection systems. Relevant codes and data are published to https://github.com/JannLi/physical_lens_soiling.

Keywords Physical optics · Soiling lens · Data generation · Image processing

1 Introduction

Autonomous driving technology relies on optical cameras to achieve reliable perception of the environment. However, cameras of autonomous vehicles exposed to various environments inevitably suffer from various abnormal situations that lead to reduced performance and reliability, such as rain, snow, fog, sand, dust, sensor fouling, and sensor damage in harsh environments. In addition to developing corresponding safety redundancy and safety actions, detecting risks in time

also reduce the risk level, according to the principle of failure mode and effects analysis (FMEA) and its risk assessment factors severity, occurrence, and detection (SOD).

Deep learning can detect various targets that are difficult to judge by artificial rules and can be applied to lens soiling detection. However, its data-driven nature brings new challenges to model training. Compared to normal working environments, the probability of obtaining corresponding various abnormal events data is low, especially in terms of camera lens soiling data. Although we can collect specific photos of mud, water mist, water droplets, and other obstructions from some autonomous vehicles, it still encounters great engineering difficulties when obtaining them on a large scale.

In solving the problem of data dependency, data generation techniques have received great attention, such as data augmentation [1, 2, 3, 4], generative adversarial networks (GAN) [5], rendering engine-based synthetic data, etc. Data augmentation techniques often use commonly used visual processing functions, which can be integrated into the deep learning model training pipeline for real-time enhancement and training. However, most data augmentation methods are unrelated to the physical optical imaging process and lack

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interpretability. GAN and diffusion models effectively generate data by simulating data distributions. However, they lack physical interpretability and depend heavily on existing datasets, making it difficult to generate data with out of bag distribution. Additionally, synthetic data generated by physical rendering engines, such as Photoshop, Blender, and UE, require additional artist costs.

Considering that types of lens soiling are infinite, from the main perspective of saving storage space, we seek for a lightweight data generation scheme that can be integrated into the deep learning online training pipeline. The method in this paper uses basic image processing functions to simulate the physical imaging process of lens soiling and designs a data generation framework based on physical optical imaging process. Firstly, we decompose the imaging process into specific steps, like occlusion, transmission, scattering, refraction, and diffusion after analyzing the physical optical imaging process of lens soiling. Then we approximate each step of the physical imaging process using basic OpenCV [6] image processing functions such as transparency addition, blurring, and grid distortion. At last we compose various effects to form the final lens soiling data generation framework, which can simulate mud stains, water stains, and water droplet refraction effects. The algorithm proposed in this paper offers physical interpretability, lightweight deployment, scalability of the framework, and compatibility with other methods through function substitution, which is a supplement to existing technologies. This paper demonstrates the effectiveness of the methods above by providing typical mud stains, water stains, droplets, and lens flare as examples, as in Fig. 1. Furthermore, the comparison of various lens soiling generation strategies is shown in Table 1. Data augmentation methods such as [7] typically add dirt by simulating noise, which does not account for the changes in light transmission through the dirt. The GAN method [8], on the other hand, can simulate more realistic effects by learning dirt patterns from existing images, but neural networks lack physical interpretability. Rendering engines like Blender, by simulating realistic light propagation, can achieve high levels of realism and physical interpretability. However, due to their slow rendering speed, they are less satisfactory in terms of deployment and online operation.

The main contributions of this paper are as follows:

- We propose a method for generating lens soiling data based on the principles of physical optical imaging, and we approximate the physical imaging process using basic OpenCV image processing functions for fast integration into deep learning training pipeline.
- We propose a lens soiling rendering framework and various effects could be generated only by adjusting functions and corresponding parameters.

- We assess the performance of our method across image classification and segmentation tasks, demonstrating its effectiveness through the results.

2 Related work

2.1 Data augmentation

Data augmentation methods process original images through basic image processing functions, with fast computation speed. These methods can be seamlessly integrated into the deep learning pipeline for real-time enhancement due to their lightweight code nature. Cubuk et al. [9] can also select from a range of data augmentation techniques based on their specific impacts on deep learning performance. Lens soiling is somewhat like data augmentation. Buslaev et al. [7] integrated many data augmentation functions, and its software package also implemented data augmentation functions for adding mud and water stains. But its generated stains are not based on physical process, and its appearance is more like data noise other than lens soiling. While conventional tools such as Photoshop can create a certain degree of dirt effects on the original image, this process heavily relies on expert-driven adjustments. We drew inspiration from the concept of data augmentation and used basic image processing functions to approximate the soiling lens effect, so that it can be integrated into the deep learning training pipeline in the future.

2.2 GAN and diffusions

Since the inception of GAN by Goodfellow et al. [5], they have been embraced across a spectrum of research domains. GAN have certain advantages in generating novel synthetic images. Jiqing et al. [10] used GAN to perform style transformation on the entire image. Zou et al. [11] analyzed the transmission and reflection effects in the physical optical imaging process of cloud layers, and ultimately got cloud layers on clear remote sensing images for generating synthetic data, while [12] used GAN for image dehazing and [13] for image deraining. Uricar et al. [14] used GAN to generate lens soiling and added it to normal images. Based on manual judgment of whether the synthesized soiling is consistent with the real soiling, the stains generated by the GAN method lack physical interpretability and require the collection of some existing data before training. Shu Y., et al. [8] incorporated physical priors by segmenting the scene into multiple layers and then applying style transformations layer by layer, which resulted in transformed images that exhibit a more pronounced sense of depth and layering. We only take the dataset [14] published for experiments. But given GAN's

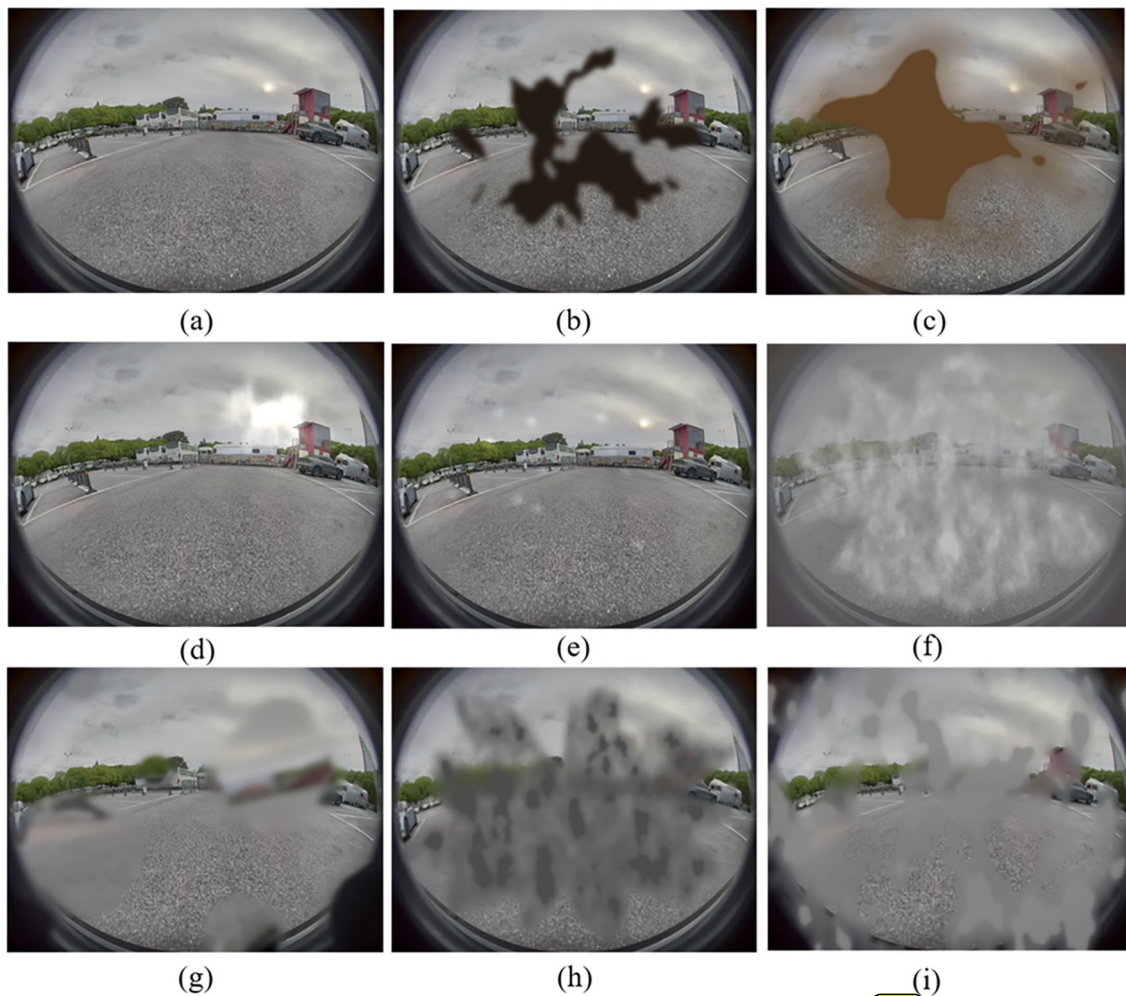


Fig. 1 Different lens soiling images generated by our method. **a** Original image; **b, c** mud stain effect; **d** noise effect; **e** dust in lens effect; **f** water stain effect; **g** water droplet effect; **h, i** water stain effect

Table 1 Comparison of various lens soiling generation strategies

Method	Interpretability	Fidelity	Automated	Online
Data augmentation	—	—	+	+
GAN or diffusion	—	+	+	o
Rendering engine	+	+	o	o
Ours	+	o	+	+

“+” represents better, “—” represents worse, and “o” represents average

impressive ability to mimic image textures, we may consider hierarchically decomposing the scene and introducing GANs to perform layer-by-layer or part-by-part transformations in the future.

2.3 Rendering engine and physics-based data generation

The advancement of computer graphics has long enabled the rendering of high-fidelity images that match physical scenes

and align with human visual perception. By simulating the process of light emission and imaging in the physical world, it generates images that closely resemble real-world visual experiences. In addition to the classic Phong model [15, 16, 17] proposed a physically based rendering (PBR) description method, which can form quantifiable and transferable rendered images. The works of [18, 19, 20] used various computer graphics (CG) production tools to produce synthetic datasets in the field of autonomous driving. We have also observed that open-source Blender software has received

great attention due to its small size, high performance, and rich open-source ecosystem. When simulating the effect of lens soiling, we used Blender software to verify the physical principles of lens distortion, soiling, and other effects. We believe that the technology based on CG may be one of the most promising methods for synthesizing data, but the learning cost of these software is slightly high. Considering the overall learning cost and software reusability, we did not use any relevant CG tools, but only referred to the optical imaging principles from it. In addition to rendering engines, other methods also process data by leveraging physical imaging processes. For example, [21] modeled image blur as a mixing process of pixels from multiple viewpoints, thereby developing deblurring algorithms. Our approach to developing algorithms based on physical laws is similarly inspired by these works. Some methods of generating foggy images from sunny images, such as the [22], analyzed the physical process of light transmission in the fog field, and correlated the transmission coefficient of light in the fog field with the pixel depth of the corresponding image, and normally the larger the depth, the smaller the transmission coefficient. However, our lens dirt appears on the surface of the lens and is independent of light transmission in the air as well as pixel depth, so image pixel depth is not taken into account.

2.4 Image composition

Image composition is the process of extracting the foreground from one image and seamlessly merging it onto another background image to create a visually coherent composite image. In a broader context, the integration of multiple visual elements from various sources into a single image is partly considered of image synthesis. Niu et al. [23] have cataloged various techniques for blending foreground images with background images. Gupta A. et al. [24] introduced a technique that involves projecting text onto the original image, followed by Poisson fusion post-processing to enhance the visual coherence. Bai et al. [25], in the domain of autonomous driving, utilized estimated ambient lighting and other parameters to feed into a rendering engine to generating a foreground image. Our approach aligns more closely with image composition, where lens soiling data are divided into the foreground soiling area and the background original image. We focus solely on processing the soiling area and then seamlessly reintegrate it with the original image.

3 Methods

3.1 Physical imaging process and simplification

Figure 2 illustrates the approximate physical imaging process of the soiling lens. The camera consists of an optical module

A. Physical Imaging Process and Simplification

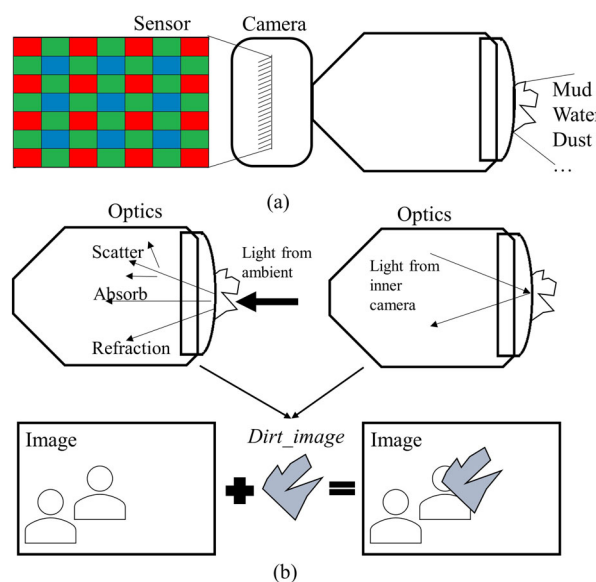


Fig. 2 Approximate physical imaging process of the soiling lens. **a** We assume the imaging process, lens soiling (mud, water, dust) happens on the lens surface. **b** Final image is composed by the original image and the dirt image, which gets light from inner camera light and ambient light

containing a lens and a sensor to which the lens soiling is attached to the external lens, as shown in Fig. 2a. The light from ambient hits the lens soiling and finally reaches the sensor through transmission, scattering, and refraction. The other part of the light comes from inside the lens, which shines from the inside of the lens to the lens soiling, and finally reflects to the sensor, and the two parts of light are combined by the sensor into the image output of the lens soiling dirt_image. The final image output is a combination of the original image and the dirty image of the lens as shown in Fig. 2b.

Taking it a step further, by utilizing a pinhole camera model, the imaging process of lens contamination can be simplified as an optical system that contains environmental scenes, lens soiling, and the pinhole camera itself.

As depicted in Fig. 3, the environmental scene is treated as a virtual light-emitting plane within the pinhole model. This plane is designed to emit rays that target the corresponding pixels of the image plane toward the optical center. In scenarios with no occlusion, these rays uniformly illuminate the pixels corresponding to the image plane, yielding photographs that are faithful to the original image. However, the presence of occlusion, in this case, lens soiling, significantly alters the imaging process within the occluded area. The imaging of this area is entirely subject to the specific optical behavior of the occlusion. Importantly, the optical

Table 2 Physical imaging process and its approximate code

Physical imaging process	Key approximate process	Key function in code
Scatter	out = blur(in)	cv2.GaussianBlur()
Refraction	out = warp(in)	skimage.transform PiecewiseAffineTransform()
Absorb	out = in $\times$ decay_ratio	In $\times$ alpha
Diffusion	out = in $\times$ lighting	base_color $\times$ light_color
Defocus blur	out = blur(in)	cv2.GaussianBlur()
Global image brightness	out = in $\times$ darkening_factor	in $\times$ dark_factor

The internal ambient light is the mean value of the external ambient light, represented as `mean(image_amb)`. The light reflected by the lens soiling onto the image plane is termed `inner_camera_color`.

$$\text{inner_camera_color} = \text{mean}(\text{image_amb}) \times \text{decay_ratio} \quad (3)$$

$$\begin{aligned} \text{dirt_diffusion_color} &= \text{inner_camera_color} \\ &\times \text{decay_ratio} \\ &\times \text{dirt_base_color} \end{aligned} \quad (4)$$

- (3) Transmission and scattering of light through lens soiling onto the image plane

The phenomenon of light scattering is effectively simulated through the application of blur, a technique that mimics the dispersion of light from a single pixel to its neighboring pixels.

We devise a clever engineering workaround for handling the varying blur effects in lens soiling images. Since OpenCV lacks a direct feature for this nuanced effect, we take a strategic approach. We start by pre-computing multiple images, each with a different level of blur. Then, for each pixel, we apply an appropriate blur effect by interpolating between these pre-computed images, tailored to its specific needs. This method uses Fourier transform principles, especially in the context of Gaussian blur. Essentially, different blur levels correspond to different retentions of the image's low-frequency details. By interpolating these details, we can achieve a desired blur effect for each pixel, creating a more natural and varied appearance in the simulation.

$$\begin{aligned} \text{dirt_scatter_color} &= \\ &\text{interpolated}(\text{blur}(\text{image_amb}, \text{coefficients1}), \dots, \\ &\text{blur}(\text{image_amb}, \text{coefficientsN}), \text{dirt_scatter_mask}) \end{aligned} \quad (5)$$

$$\begin{aligned} \text{dirt_scatterTrans_color} &= \text{dirt_scatter_color} \\ &\times \text{transparent_mask} \end{aligned} \quad (6)$$

- (4) Transmission and refraction of light through lens soiling onto the image plane

Lens soiling from water droplets typically causes irregular distortions due to light refraction. To simulate this, we use the grid distortion technique as referenced in [26], which created an effect like the liquefaction tool in Photoshop.

$$\begin{aligned} \text{dirt_refraction_color} &= f_warp(\text{image_amb}, \\ &\text{warp_roi_mask}) \end{aligned} \quad (7)$$

$$\begin{aligned} \text{dirt_refracTrans_color} &= \text{dirt_refraction_color} \\ &\times \text{transparent_mask} \end{aligned} \quad (8)$$

In terms of grid distortion code implementation, we apply the image processing library `skimage.transform` [27]. And in our paper, polygons with different scales in the same center are generated as control points to achieve the distortion effect. Results are demonstrated in Fig. 5. The specific process of simulating lens refraction by image grid distortion (taking the image enlargement effect as an example) is illustrated Fig. 4.

- (5) Lens defocus: The lens soiling object is attached to the lens which is out of focus. [7] implemented the lens defocus effect function, and Gaussian blur could also be selected to mimic lens defocus function.

$$\text{dirt_image} = \text{defocus_blur}(\text{dirt_image}) \quad (9)$$

- (6) Mask generation with random shapes: To handle various lens soiling effects like dirt, transmission, refraction, and scattering, we need specific masks. While it is feasible to gather these masks from artistic designs online, our approach is more systematic. In this paper, we employ a random function to procedurally generate these masks. We utilize the `PythonPerlin` [28] library, which harnesses Perlin noise to create random shapes. This method not only produces effects like clouds and irregular forms but also allows us to blend various noise textures, crafting unique masks for our simulations.

Fig. 5 Water droplet enlargement effect by grid distortion



- (7) Global image brightness adjustment: When the lens is occluded, the effective aperture of the camera becomes smaller, resulting in a weaker overall light intensity in the captured image, similar to the effect of reducing the camera's aperture. To simulate it, we decrease the overall brightness of the image. The reduction factor is determined by the ratio of the occluded area to the total area of the image.

$$\text{dark_factor} = f\left(\frac{\text{area}(\text{dirt_image})}{\text{area}(\text{image})}\right) \quad (10)$$

$$\text{image} = \text{image} * \text{dark_factor} \quad (11)$$

After decomposing the imaging process into basic image process functions, we find that almost all hyper parameters in the functions could be represented through random shapes. Main work of improving the method turns to fetching more random shape generation methods.

$$\text{x_mask} = \text{generate_masks}() \quad (12)$$

The final total rendering equation is:

$$\begin{aligned} \text{image} = & (\text{image_amb} \times (1 - \text{dirt_mask}) \\ & + \text{defocus_blur}(\text{dirt_image}) \\ & \times \text{dirt_mask}) * \text{dark_factor} \end{aligned} \quad (13)$$

where the shape of lens soiling is set as dirt_mask. If there is no refraction and no scatter:

$$\text{dirt_image} = \text{dirt_diffusion_color} \quad (14)$$

If only no refraction:

$$\begin{aligned} \text{dirt_image} = & \text{dirt_diffusion_color} \\ & + \text{dirt_scatterTrans_color} \end{aligned} \quad (15)$$

If there is refraction:

$$\text{dirt_image} = \text{dirt_diffusion_color} + \text{dirt_refracTrans_color} \quad (16)$$

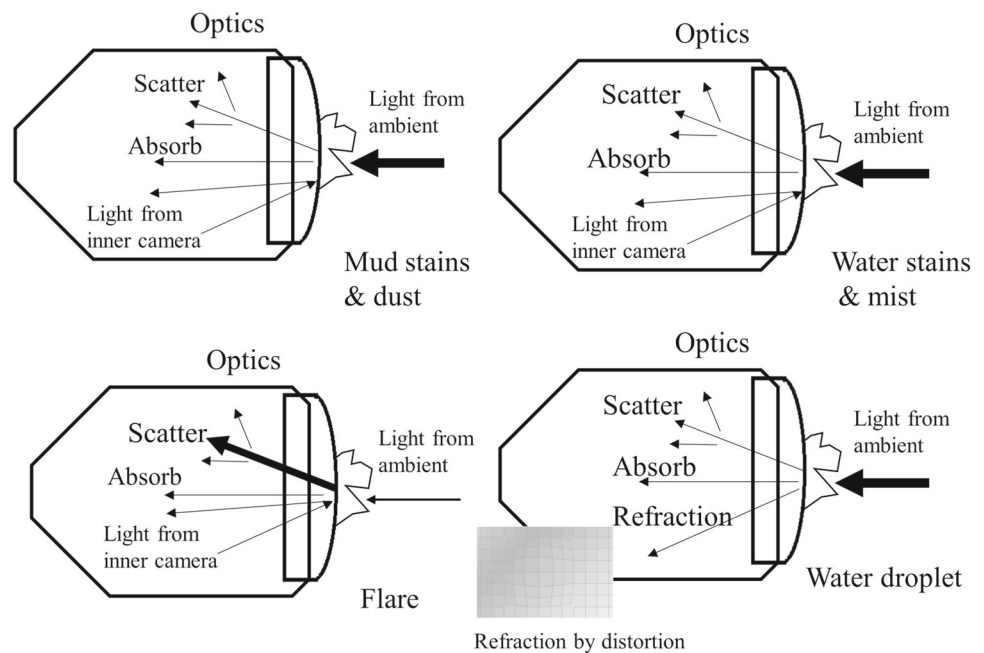
3.2 Certain lens soiling data generation

We create effects like mud and water stains, as well as water droplet refraction and lens flare, using our established framework. To generate these effects, we just need simply modify the parameters in our rendering equations and apply minor refinements. The process seems like a mini rendering engine.

For mud stains and dust, external light refraction and scattering are not considered. For water stains and mist, the water is assumed to be either dirty or thin, so refraction is neglected, and its color is typically white or gray. For water droplets, light refraction is accounted for using grid distortion to approximate the effect. For lens flare, external light is magnified and blurred based on the shape of the dirt.

Empirically, water vapor and dirt on lenses and eyeglass lenses typically accumulate more at the edges. This is also true for fogging on the inside of car windshields, which usually begins at the edges. However, upon examining the actual lens dirt datasets Woodscape [14], we found that the distribution of dirt is random. We speculate that this is because the water vapor and dirt on lenses, eyeglass lenses, and car windshields originate from condensation of internal chamber moisture. Since the temperature at the lens edges differs from that at the center, condensation starts at the edges. In contrast, the soiling lens datasets consist of images captured by fisheye cameras mounted on the exterior of cars, where the dirt originates from environmental splashes, such as rainwater and mud. Therefore, the distribution of dirt is more random. Although there is no theoretical basis for this, we adopted a random dirt distribution in our model, consistent with the datasets. Our physical model and algorithmic framework support various dirt distributions by modifying the mask shape.

Fig. 6 Physical simplification of different lens soiling effects



1. Mud stains: As mud stains in Fig. 6, external light refraction and scatter are not considered. The mud color is dark with low light transmittance. Formulas (13) and (14) are used and for `dirt_base_color` in formula (4), we use a dark color with random jitter. Fine-tuning the mask thresholds completes the process.
2. Water stains: As water stains in Fig. 6, the same as mud stains, external light refraction is not considered and we refer to formulas (13) and (14) for calculations. For `dirt_base_color` in formula (4), water color tends to be white or gray.
3. Water mist: The same as water stains with different color and threshold of transparent masks.
4. Water droplet: As water droplet in Fig. 6, light refraction is taken into account. We choose formulas (13) and (15) to fine-tune the shape and position of the `warp_roi_mask`. Each water drop effect is crafted individually and the process is completed by fine-tuning the mask thresholds.
5. Lens flare: As lens flare in Fig. 6, flare arises from excessive light and is handled similarly to water mist effects utilizing formulas (13) and (14). The main difference lies in the adjustment of the `dir_mask` to ensure the mask covers more than a single pixel (mask area factor > 1) while ensuring image values are capped at 255.
6. Dust in lens: As dust in Fig. 6, dust within the lens seems like mud stains but receiving more ambient light, appears lighter. We employ formulas (13) and (14) for it. The dust shape is created using a `dirt_mask`, following a process akin to that for mud stains. Transparency mask value is increased and image values are ensured to be capped at 255.

Figure 1 shows examples of different types of dirt described above.

7. Data label generation: We primarily utilize a combination of `transparent_mask` and `dirt_mask` to create labels, as depicted in Fig. 7b. When saved in PNG format, the transparency values span from 0 to 255. To differentiate between clean, transparent, semi-transparent, and opaque areas, we apply various thresholds. For instance, in the case of mud, the ranges might be set as 0 ~ 20 for clean, 20 ~ 150 for semi-transparent, and 151 ~ 255 for opaque.

4 Experiments

To validate the effects of our synthetic data, besides comparing with real lens soiling data, we also conduct experiments based on the soiling parts. For a fair comparison with real lens soiling data, we use similar amount of synthetic data as real ones.

4.1 Dataset and method

We take the Woodscape [14] as our benchmark dataset, which is a real fisheye dataset collected by vehicles, including both clean and soiling fisheye lens images. Clean images consist of 6185 training images and 2049 validation images, while soiling images include 4000 training images and 1000 test images accompanied by manually annotated soiling semantic segmentation masks, categorized into four classes: clean, transparent, semi-transparent, and opaque as illustrated in

Fig. 7 **a** Is generated directly by PythonPerlin. **b** is obtained by multiplying **(a)** and an irregular polygon

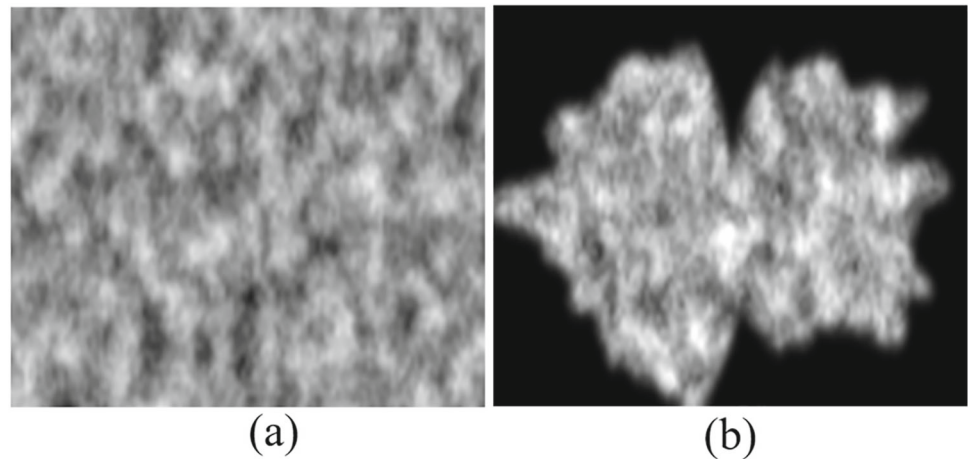


Fig. 8 Two images and corresponding labels from dataset [14]. **c**, **d** are display the labels for **a**, **b**. Within the labels, black denotes “clean,” green signifies “transparent,” “blue” indicates “semi-transparent,” and red represents “opaque”

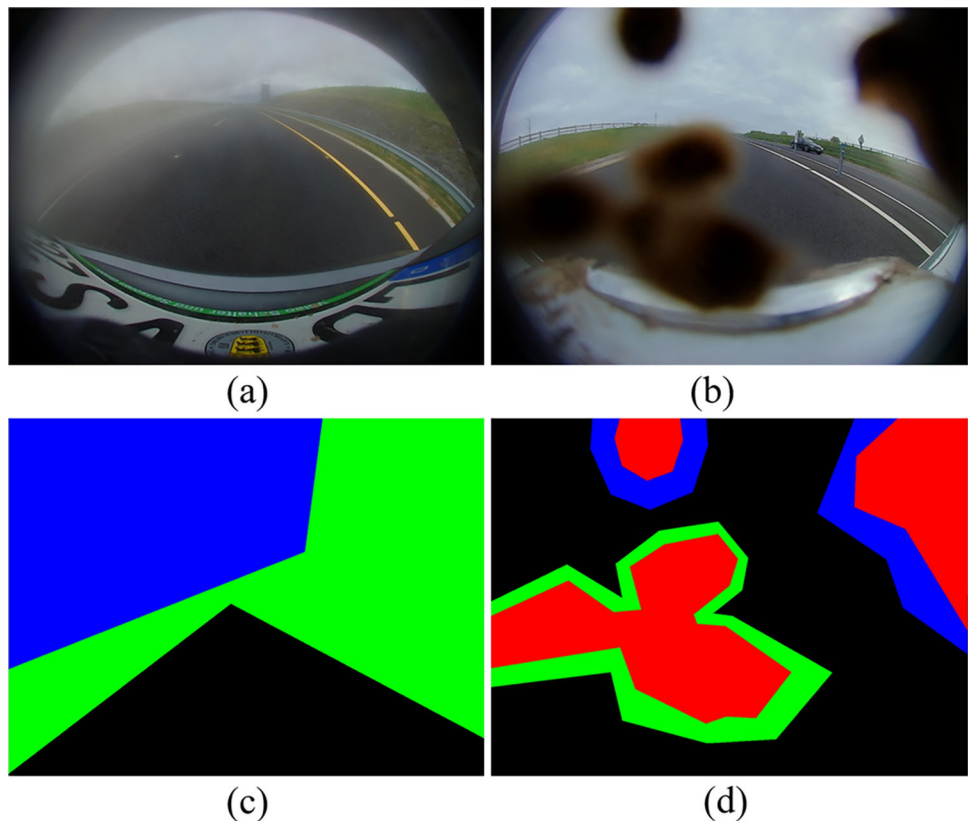


Fig. 8. Based on our observations, due to the highly subjective nature of manual annotations in judging the degree of soiling, we merge the categories by combining the transparent category with the clean category, then synthesize soiling fisheye images on clean ones with method mentioned above, and set a mask threshold to divide them into three categories. Examples are presented in Fig. 9.

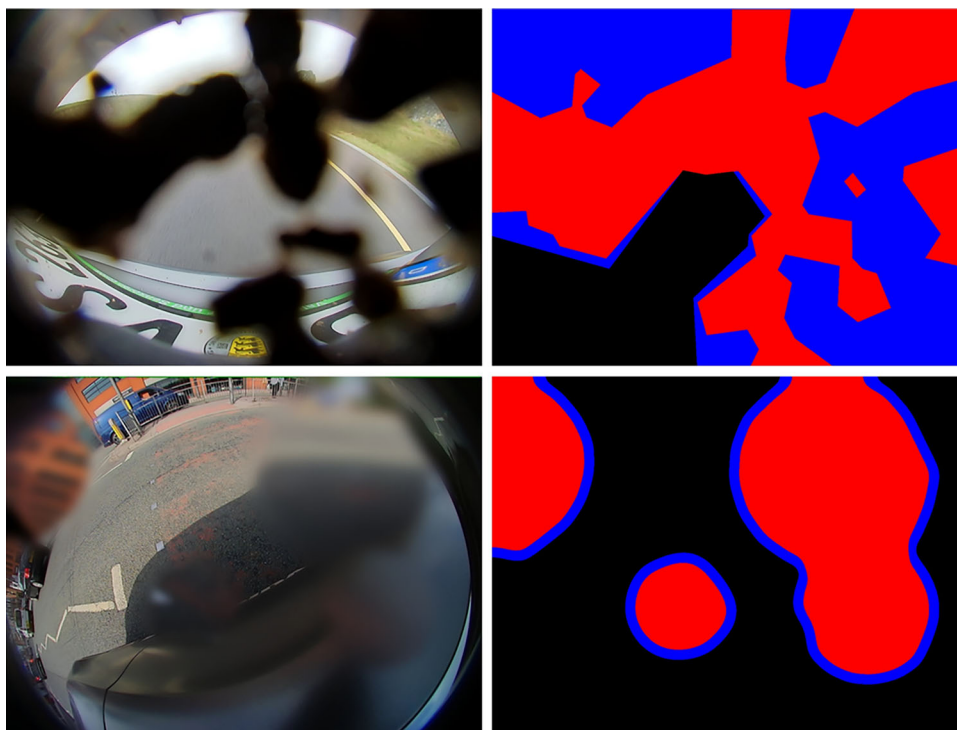
We employ the recently highly acclaimed large model InternImage [29] as our benchmark model. It can perform both semantic segmentation and image classification tasks.

4.2 Experimental settings

We synthesize soiling images on authentic clean images, comprising 6185 training images and 2049 validation images, each augmented with one type of soiling such as mud stains, water mist, or water droplets.

In the image classification experiment, we construct training sets by mixing real soiling images with real clean images. We then replace the soiling images with synthetic ones or use a combination of both. The test data consists of a mix of real soiling and real clean test images. We perform binary

Fig. 9 Top: real soiling image and its soiling semantic segmentation mask. Bottom: synthetic soiling image and its soiling semantic segmentation mask



classification between soiling and clean images, using top-1 accuracy as the evaluation metric.

In the semantic segmentation experiment, we use a validation set of 1000 real soiling images as the test set, and train the model with real soiling training images, synthetic soiling training images, and a mixture of both as the training set, respectively. The final test mean intersection over union (mIoU) is used as the evaluation metric.

5 Results and discussions

5.1 Generated data

We present our results using a sample image from [14], as shown in Fig. 10. This figure illustrates the original image alongside its corresponding soiling masks for water droplets, mud, and water stains. These soiling masks are categorized based on opacity levels.

Compared to manual labeled mask in Fig. 9, labels from synthetic images are quantitative and accurate. In addition, the manual judgment of transparent, semi-transparent, and opaque areas in real data is also lack of quantification and unstable.

We evaluated the generation speed of various lens soiling effects by creating 100 images with completely random masks. We conducted the test on a CPU-based host with an Intel(R) Core(TM) i9-9880H CPU, which features 8 cores and 16 threads, and has a base frequency of 2.4 GHz. The

Table 3 Comparison of generated speed of different lens soiling. The unit of measurement is seconds

Method	Time/s
Mud	0.131
Water stains	0.163
Flare	0.142
Water droplet	24.3

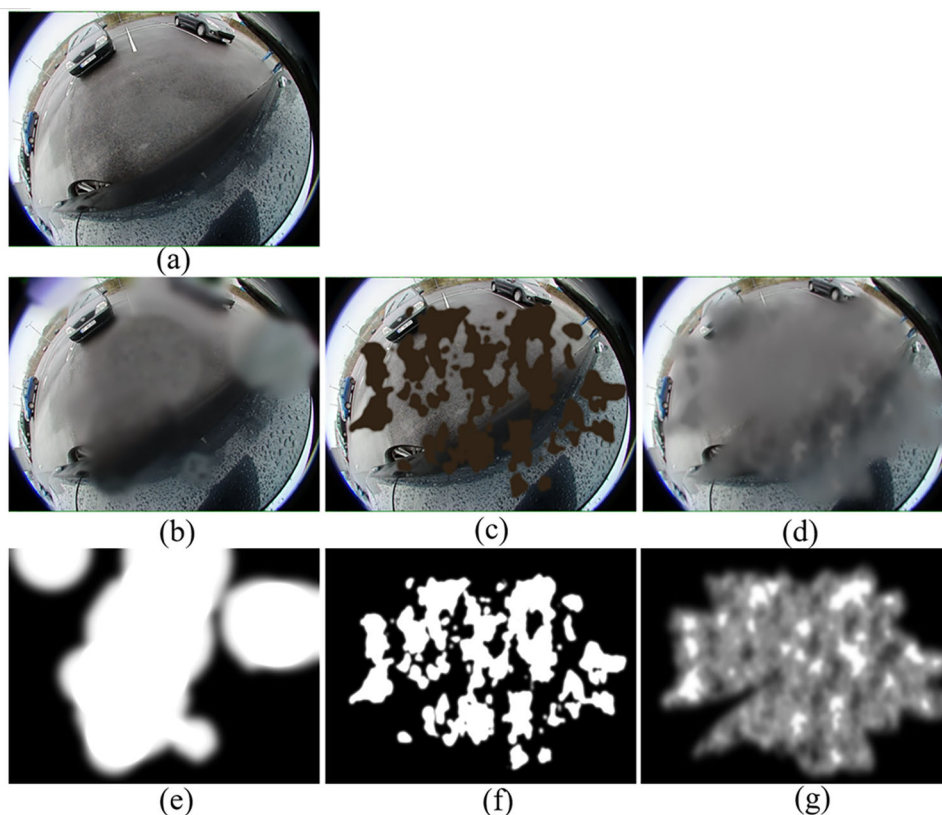
test was performed using single-threaded mode. The findings are summarized in Table 3. Although additional work is required to refine the water droplet function, integrating functions for flare, mud stains, and water stains into an online deep learning training pipeline is feasible.

5.2 Performance of generated data in downstream tasks

(1) Image classification

We conduct image classification experiments in the manner described above and results are shown in Table 4. The model trained with synthetic data demonstrates performance comparable to that of the model trained with real data. When we combine two parts of training data, the top-1 classification accuracy even increased by 1.58% compared to that of real data. It is noteworthy that our synthetic data are generated based

Fig. 10 Generated examples. **a**: The original image. **b,e**: The water droplet image and corresponding mask. **c,f**: The mud stains and corresponding mask. **d,g**: The water stains and corresponding mask



on real clean data, which means the domain of synthetic data is very similar to that of real clean data. However, real soiling data and real clean data are not from the same batch, and there is some difference between them. Therefore, the difficulty of classification using synthetic soiling data is greater, which further highlights the value and authenticity of our synthetic data.

For comparison, we also use alumentations [7] and CycleGAN [14] to synthesize a batch of dirty images of the same quantity for image classification model training, respectively. The results, as shown in Table 4, indicate that our method is close to CycleGAN and is stronger than the data augmentation method. At the same time, the comparison of effects shown in Fig. 11 also reveals that alumentations can only add dirt by uniformly adding noise, which is much less realistic. In contrast, CycleGAN, which learns from real images, achieves better integration of the synthesized dirt with the background image. However, the inference speed of

CycleGAN on our CPU exceeds 2.5 s per frame, which is more than ten times slower than our method. Additionally, our method does not require pre-training with pre-collected data.

(2) Semantic segmentation on soiling part

We categorize “semi-transparent” as “clean” and conduct experiments with only two types of category segmentation, and results are shown in Table 5. The model trained solely with synthetic data could achieve approximately 63% mIoU as model trained with real data, which is like the results obtained with GANs as reported in [14]. However, when we mix the two sets of data, there is a certain improvement in the model’s recognition ability (+ 0.49% on mIoU). This indicates that our synthetic data have a positive effect on enhancing the model’s performance. Nevertheless, the effect is not very noticeable due to the difference in the manual annotation of the original data. The segmentation examples are shown in Fig. 12.

Table 4 The acc@1 of image classification on soiling images and clean images with different soiling data generation methods

Training data	Original images	Alumentations	CycleGAN	Ours
Real	93.70%	–	–	–
Synthetic	–	66.78%	93.64%	92.10%
Mixed	–	86.22%	96.87%	95.28%

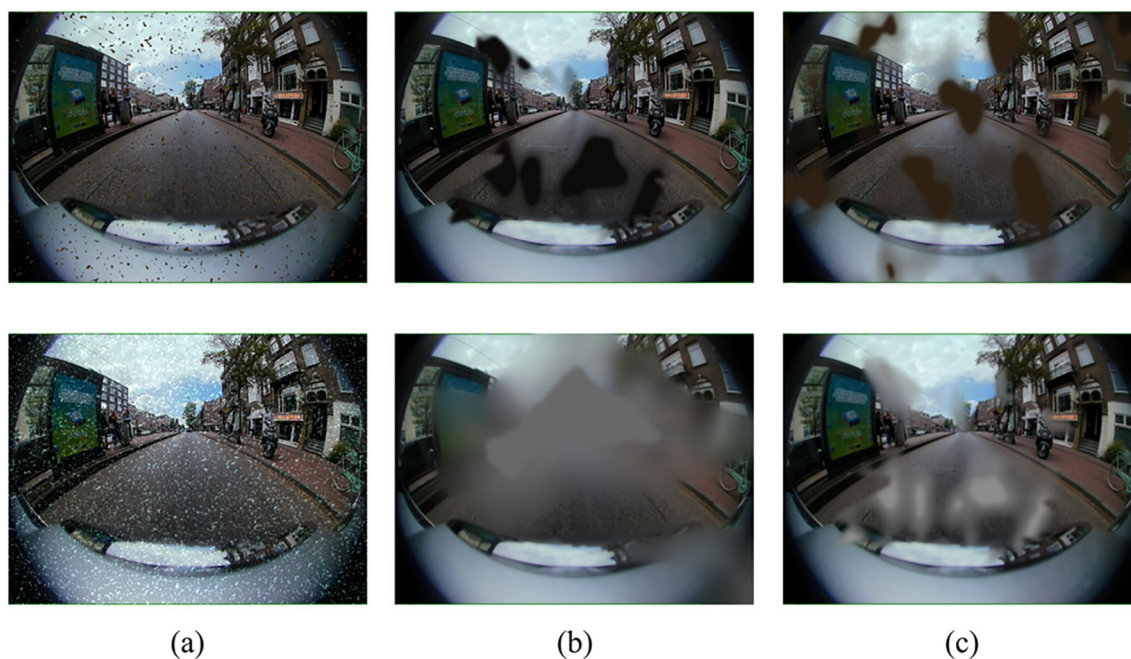


Fig. 11 Comparison of the effects of synthetic data from different methods. Upper: Mud stains; Below: Water stains. **a** Albumentations; **b** CycleGAN; **c** Ours

Table 5 The results of soiling semantic segmentation in test data of real soiling images with two classes: clean & opaque

Training data	Testing data	Clean	Opaque	mIoU
Real	Real	95.7	90.53	93.11
Synthetic	Real	61.0	56.63	58.81
Mixed	Real	95.95	91.25	93.6

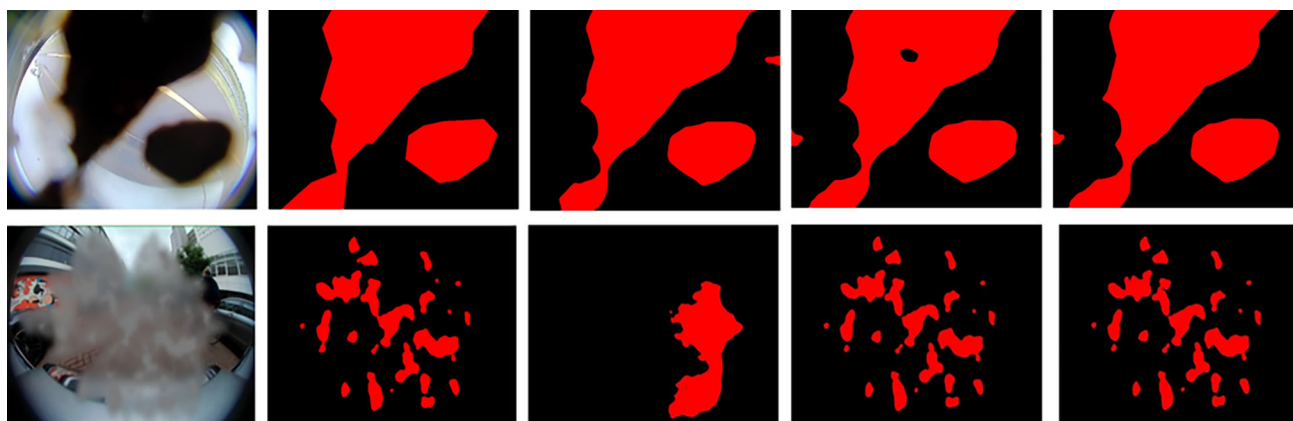


Fig. 12 Examples of segmentation results with two categories. Top: Results of a real soiling image. Bottom: Results of a synthetic soiling image. From left to right: RGB images; ground truths; results trained with real data; results trained with synthetic data; results trained with mixed data

5.3 Ablation study

We conduct comparisons to assess the impact of eliminating the aforementioned physical processes and validate their effects on the image classification task, as detailed in Table 6. After removing certain processes, the accuracy decreased to

varying degrees. In particular, the removal of ambient lighting led to a significant drop in accuracy, down to 71.22%. As discussed in last chapter, ambient lighting exclusively illuminates lens soiling, which makes it more harmonious with the surrounding environment. Therefore, it holds a more significant position in the synthetic lens soiling data when compared

Table 6 The acc@1 of image classification on soiling images and clean images without certain physical process. Only synthetic data are used for training. We add random jitter to the base dirt color, so in “Constant base color,” we remove the random jitter

Physical process	acc@1(%)
Constant base color	81.26
Without ambient lighting	71.22
Without inner camera lighting	87.31
Full	92.10

with inner camera lightning, which controls the brightness of lens soiling. The consequences are depicted in Figs. 13 and 14.

In fact, the mud and rain effects presented in albumenations [7] can be regarded as simplified versions of our processes. Moreover, they not only lack the geometric complexity of lens soiling, but also fail to account for the physical phenomena of camera inner lighting and ambient lighting interactions (scatter).

Furthermore, we compare the effects of different data quantities on segmentation accuracy in Woodscapes [14] datasets, the number of real training set was 4000, and the

Table 7 The results of soiling part segmentation with different quantities of synthetic training images

Quantities	Clean	Opaque	mIoU
57,638	59.13	23.98	41.56
30,736	61.0	56.63	58.81
14,030	56.48	10.84	33.66
3723	54.8	2.97	28.89

experiment shows that the best effect is achieved when the amount of synthetic data is set 30,736. The results are shown in Table 7.

5.4 Additional topic

Table 8 illustrates the semantic segmentation results with 3 classes: clean, semi-transparent, and opaque. The results indicate that models trained exclusively on synthetic soiling data struggle to recognize the “semi-transparent” and “opaque” categories in the test set. This difficulty likely stems from the fact that the test set labels are manually annotated without strict quantitative standards for defining category boundaries, resulting in substantial discrepancies from our

Fig. 13 Mud stains without certain physical process. **a** Original image; **b** mud stains effect with full process; **c** mud stains without inner camera lighting, its color looks brighter; **d** mud stains without ambient lighting, just mud’s base color

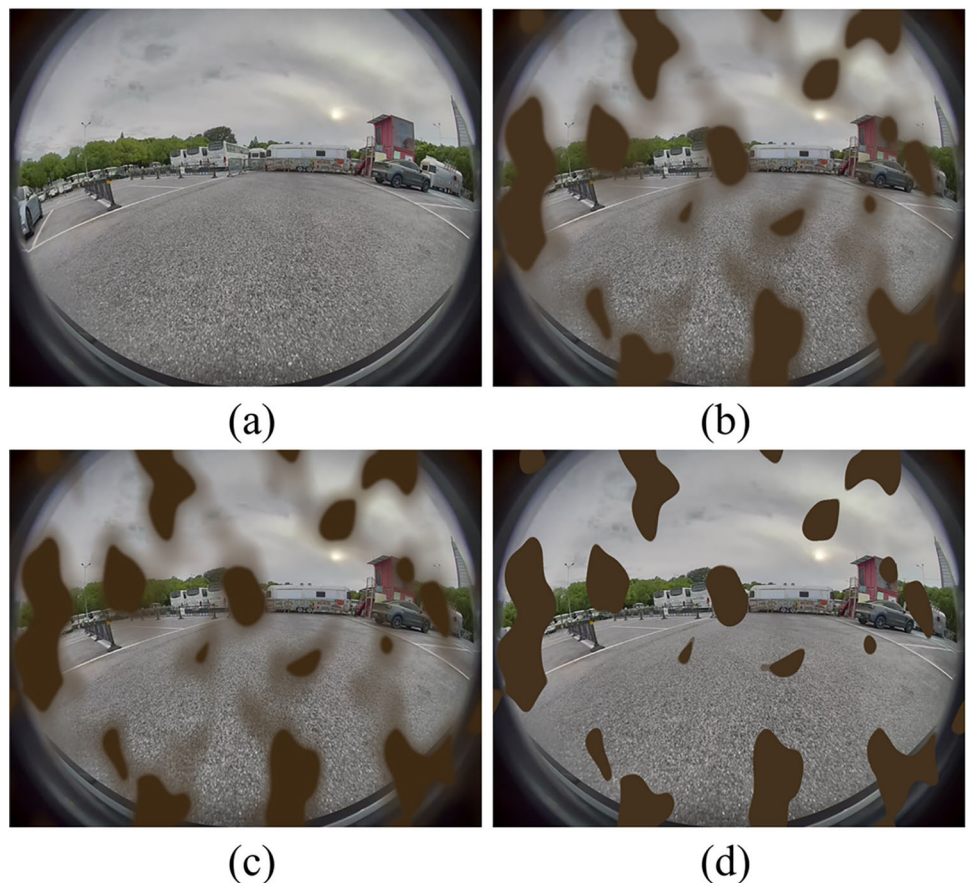
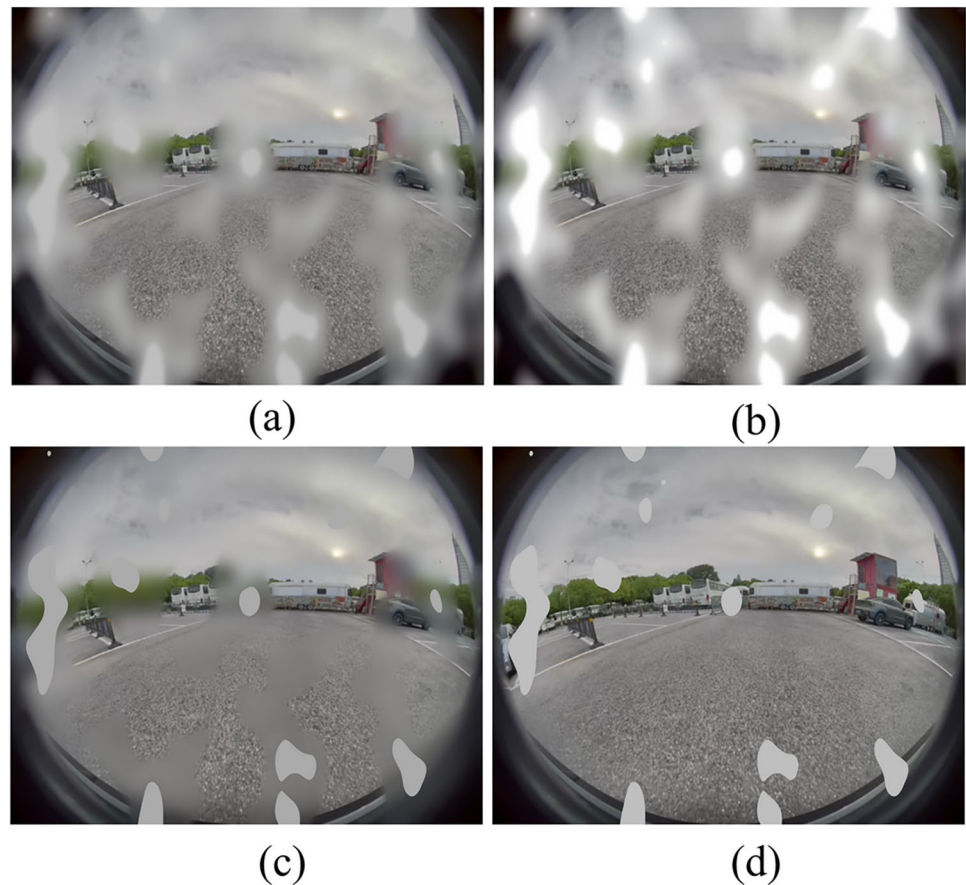


Fig. 14 Water stains without certain physical process. **a** Water stains effect with full process; **b** water stains without inner camera lighting, its color looks brighter; **c** water stains without ambient lighting's absorb, only scatter; **d** water stains without ambient lighting's scatter and absorb



standardized labels. Additionally, we observe that models trained on real data also encounter challenges when classifying “semi-transparent” and “opaque” categories on synthetic test images. However, models trained on synthetic and mixed datasets exhibit significantly enhanced recognition capabilities, suggesting that the integration of diverse data sources can effectively improve model performance in this context. There is still a domain gap between synthetic and real images; we will explore more solutions such domain adaption and domain randomization in future work to eliminate it.

Figure 15 displays some of the segmentation results. In comparison with the experiment detailed in Table 5, it validates the advantage of precise labeling by synthetic data.

6 Conclusion

In this paper, basic image processing functions are used to approximately simulate each physical step of lens soiling imaging. Various effects are composed to form the final lens soiling data, which can simulate the effects of mud, water stains, water drop refraction, flare, and so on. The generated data demonstrate promising effectiveness for soiling image classification on a real-world public dataset. Although the results for soiling semantic segmentation are not fully satisfactory due to non-standard manually annotated labels, category adjustment still leads to improved outcomes. The generated data could be used for lens soiling classification

Table 8 The results of soiling semantic segmentation in test data of real soiling images with three classes: clean & semi-transparent & opaque

Training data	Testing data	Clean	Semi-transparent	Opaque	mIoU
Real	Real	94.45	76.41	92.81	87.89
	Synthetic	80.58	29.01	47.71	52.43
Synthetic	Real	55.85	0	0.09	18.62
	Synthetic	96.52	86.13	93.81	92.15
Mixed	Real	92.34	43.75	52.52	62.87
	Synthetic	97.14	87.73	94.32	93.06

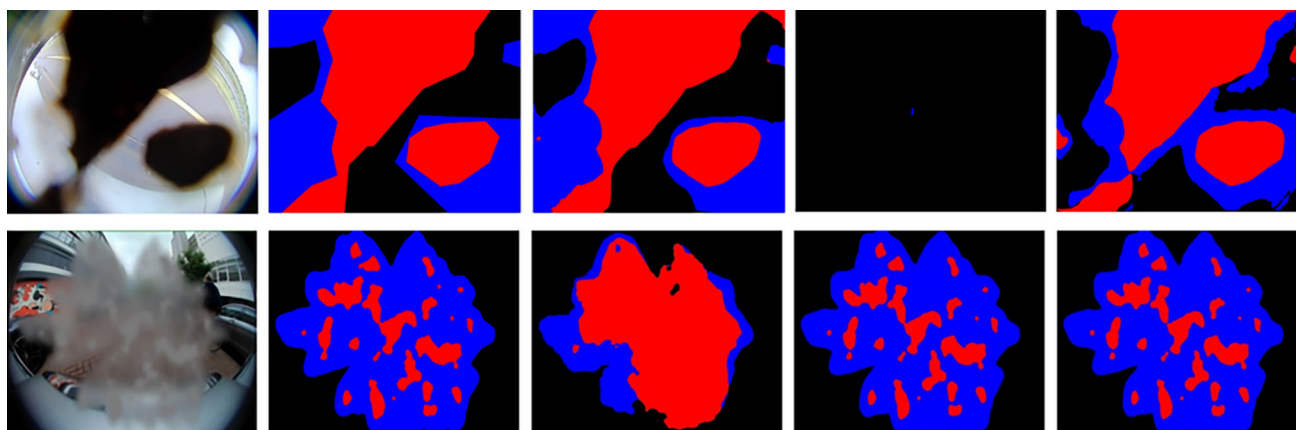


Fig. 15 Examples of segmentation results with 3 categories. Top: Results of a real soiling image. Bottom: Results of a synthetic soiling image. From left to right: RGB images; ground truths; results trained with real data; results trained with synthetic data; results trained with mixed data

task and a warm start for semantic segmentation task. The approach utilizes fundamental OpenCV image processing functions, which can be seamlessly integrated into the deep learning training pipeline.

This approach offers significant advantages over manually collected labeled data, not only facilitating easy data acquisition but also eliminating the costs associated with labeling. The resultant soiling labels are more accurate and controllable, yielding image quality that surpasses that of data augmentation methods, as evidenced by the positive outcomes in real-data experiments.

The introduction of large language models can help alleviate the problem of data scarcity, so they are becoming more and more widely used in the field of autonomous driving [23–32], and we may also introduce large language models to solve the problem of lens soiling in the future. In addition, the domain generalization problem of synthetic and real data needs to be solved, so we can also use methods such as domain adaptation [30] to improve the efficiency of our synthetic data in future work.

Author contributions Song Zhang and Rui Zhang came up with the idea and the main theory of the method, and they wrote the main part of the paper. Jian Li conducted experiments to validate the effects of the method. Xuefeng Li reviewed the manuscript and gave advice on modifying the method.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflicts of interest.

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